Prem **Mallappa**

Prem Mallappa - CV (2-page) Systems Software Architect • Embedded Systems • Virtualization

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Seasoned Software Architect with extensive expertise in designing and developing high-performance embedded systems, system software, Linux kernel drivers, and virtualization technologies. Proficient in a wide range of programming and scripting languages, with a proven track record of delivering robust, scalable solutions in complex technical environments.

A highly analytical and collaborative team player, recognized for strong problem-solving abilities, logical thinking, and a passion for mastering emerging technologies. Committed to driving innovation and excellence through clean architecture, optimized code, and best-in-class engineering practices.

Skills

Leadership	**22+ years** of multidisciplinary team and project leadership experience; **software architecture** and technical direction across systems and platform teams.
Communications	**Clear written and spoken communication** for architecture reviews, design discussions, mentoring, and cross-functional execution.
Technology	Broad and deep IT expertise across **embedded systems**, **system software**, **cloud/security**, **virtualization**, and software services development.
Computer Security	**Cryptography**, virtualization and cloud computing security architecture, **secure software development**, risk management, compliance, intrusion detection, and prevention.
Programming Languages	**C**, **C++**, Assembly (**x86**, **ARM**, **MIPS**), Rust, Python, Go, BASH
Operating Systems	**Linux**, FreeBSD, Symbian
RTOS	**VxWorks**, **QNX**, seL4, Zephyr, ThreadX
Architecture	**x86**, **ARM v6/v7/v8**, **MIPS64**, PowerPC, RISC-V
Development Tools	**GCC**, **Clang**, **Git**, **CMake**, Make, GDB, Perf, VTune, ARM Development Tools, Keil, JTAG, Docker, QEMU
Systems & Virtualization	**Linux kernel development**, **device drivers**, hypervisors, **IOMMU/SMMU**, pre-silicon validation, boot flows, and low-level firmware bring-up.
Performance & Acceleration	**SIMD/vectorization**, **performance optimization**, math and crypto libraries, **SmartNIC/DPU offload**, NVMe-oF, RDMA, and hardware-assisted data paths.
Specialized Skills	Embedded systems, system software, **cryptography**, filesystems, performance engineering, packet/data-path acceleration, and platform enablement.

Education

Masters (Computer Science)
Pilani, Rajasthan, India

BITS Pilani — *2016*

Bachelors (Computer Science)
Chikkmagalur, Karnataka, India

Adichunchangiri Institute of Technology (AIT) — *2002*

Affiliated: Visvesvaraya Technological University (VTU)

Open Source

QEMU (Link) *2014-2016*

- SMMUv3 (IOMMU) emulation support for ARMv8
- Designed and implemented SMMUv3 model merged into mainline
- Added support for Stage1, Stage2, and nested virtualization
- Implemented command queue processing and page table walk

Linux Kernel (Link) *2011-2016*

- MIPS Kexec/Kdump port and IOMMU subsystem
- Developed MIPS64 port of Kexec and Kdump (merged upstream)
- Fixed critical bugs in KEXEC for Cavium Octeon platforms
- Contributed to IOMMU/SMMUv3 driver development

GLIBC (Link) *2018-2020*

- Performance optimizations for AMD processors
- Fixed memcpy behaviour on AMD processors
- Optimized string functions for x86\\$_{64}$ architecture
- Performance improvements for memory operations

AMD LibM (Link) *2018-2025*

- Open source math library for AMD processors
- Core contributor to open sourcing AMD Math Library
- Optimized transcendental functions (exp, log, pow, trig)
- Implemented SIMD/FMA optimizations for vector operations

Experience

AMD Ltd.

2018 – Presen

Bengaluru, India

PRINCIPAL ENGINEER

Worked in multiple teams within AMD including **AMD LibM**, **AOCL Cryptography**, **BSP**, and **NVMe** in **Pensando** BU. Every project was different, requiring quick learning and adaptation to new domains. The role also involved **leading a team** of four-to-six engineers in various projects. I was also part of the leadership team where I mentored engineers from across AOCL and Pensando BU.

SR. MEMBER OF TECHNICAL STAFF

Worked on **Model0** which was **pre-silicon validation** of next generation AMD cores, including Zen2 and Zen3. The role involved working with hardware team to bring up bare-minimum processor configuration with just core and memory controller, and run Linux, stressors and benchmarks. This involved working on low-level firmware, bootloaders, Linux kernel and performance analysis tools.

Broadcom Ltd.

2014 – 2016

Bengaluru, India

PRINCIPAL ENGINEER

Architected foundational software stack for Broadcom Vulcan, a multicore-multithreaded ARMv8 64-bit server processor. Engineered complete **SMMUv3** (System Memory Management Unit) driver infrastructure supporting advanced IOMMU virtualization. Developed and contributed an **SMMUv3 emulation model** to QEMU mainline, enabling early platform validation and ecosystem enablement. Implemented comprehensive virtualization features including command queue processing, STE/CD descriptor parsing, multi-level page table walks, and stage-1/stage-2 address translation.

Cavium India Pvt. Ltd.

2011 – 2013

Bengaluru, India

TECH LEAD

Led low-level software development for Cavium Octeon III (MIPS64) multicore network processors, delivering stable platform firmware for networking customers. Resolved a critical KEXEC kernel bug and authored the full **MIPS64 port** of the Kexec/Kdump crash dump infrastructure, ultimately merged into Linux mainline. Engineered a bare-metal core dump generation framework with a host-resident daemon that enabled detailed post-mortem GDB analysis during bring-up. Designed and implemented the **CavHv hypervisor** for MIPS64 architecture using hardware virtualization extensions to validate next-generation features.

ARM Ltd.

2005 – 2009

Bengaluru, India

SR. DEVELOPMENT ENGINEER

Executed comprehensive **OS porting initiatives** for emerging ARM architectures including ARM1176JZFS with TrustZone security extensions and the Cortex-A8 application processor. Developed a production-grade touchscreen driver for Symbian OS along with a Python-based automated validation framework used across multiple programs. Designed a precision interrupt latency measurement infrastructure that quantified TrustZone secure-world context switching overhead for silicon partners. Ported the **L4Ka::Pistachio microkernel** to ARM architecture, enabling virtualization research and multiple proof-of-concept implementations.

Sasken Communication Technologies Ltd.

2004 – 2005

Bengaluru, India

SR. SOFTWARE ENGINEER

Architected and maintained the **Extended File System (EFS)** for VxWorks-based UMTS base station infrastructure, balancing throughput with resiliency requirements. Engineered a reset-resilient filesystem with **advanced wear-leveling algorithms** so data integrity held across unexpected power cycles in the field. Designed a flash-optimized flat file tree structure that minimized write amplification and significantly extended device longevity.

Global Edge Software Ltd.

2003 – 2004

Bengaluru, India

SOFTWARE ENGINEER

Designed and implemented a **comprehensive SDIO host controller driver** for Linux on Intel StrongARM embedded platforms used in consumer devices. Engineered a high-performance 4-bit SDIO driver for Marvell 802.11g WiFi chipsets that consistently achieved maximum theoretical throughput. Optimized **DMA transfers** and interrupt handling paths to minimize latency and maximize wireless data transfer efficiency under heavy load.

Short Stints

VSPL Ltd.

Software Architect

Nov. 2016 – Jan. 201

Bengaluru, India

Cisco Ltd.

Software Engineer

Oct. 2010 – Apr. 201

Bengaluru, India

B-Labs, London UK

Sr. Engineer - Contractor

Nov. 2009 – Sep. 201

Bengaluru, India

Harman International

Engineer

Jul. 2009 – Nov. 200

Bengaluru, India